Spokes: Reproducible Board Reporting for The Northwest Hub

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Live since Jun 2026
Lightspeed KPI sync operational

Hourly	2,736	90 / 90	Live
KPI Sync	**Rows Loaded**	**Day Coverage**	**Dashboard**
Lightspeed · Current	Neon Postgres warehouse	Lightspeed · 0 days stale	Grafana on Fly.io · Jun 2026

Lightspeed API ▶ GitHub Actions ETL ▶ Neon Postgres ▶ Grafana on Fly · *QBO integration in progress · Lightspeed slice operational · GoogleSheets postponed*

① Introduction & Motivation

Problem. The Northwest Hub manages financial reporting across Google Spreadsheets, QuickBooks Online (QBO), and Lightspeed R-Series POS. Leadership reporting relied on manual exports, spreadsheet reconciliation, and ad-hoc board packets.

Research question: How can The Northwest Hub run a low-cost, reproducible reporting workflow, and what do trailing 90-day KPI and directional-change analyses reveal about recent performance?

Fig 1. Pipeline architecture. Source systems feed scripts, scripts load Neon warehouse tables, and Grafana serves board-facing reporting from a

Stakeholders.

- Board members
- Executives
- Staff

③ Results & Discussion - Grafana board snapshot. The live NDahs board includes (1) trailing 90-day KPI trend panels, (2) directional change panels for prior-window and year-over-year comparison, and (3) freshness/coverage signals used for claim boundaries. Refresh cadence is hourly for Lightspeed-fed panels; QBO-linked coverage remains flagged in progress.

② Data & Pipeline

Data sources. Lightspeed R-Series POS (live hourly sync) and QBO (waiting on secrets).

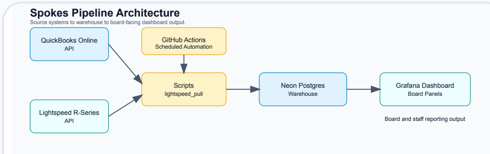
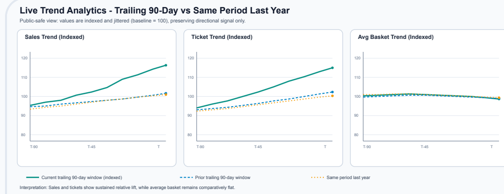


Fig 1. Pipeline architecture. Lightspeed feeds ETL scripts scheduled by GitHub Actions. Normalized records land in Neon Postgres; Grafana on Fly serves board-facing panels. `financial_balances` dominates (2,742 rows as of 2026-07-31), reflecting 15 months of hourly Lightspeed snapshots after backfill through 2024–2026.

Pipeline highlights: UPSERT loading (idempotent inserts), refresh-token rotation persisted to `oauth_tokens`, and dedup check script post-backfill.

③ Results & Discussion

Live Trend Analytics: Trailing 90-Day vs SPLY. Analysis answer: indexed trends show strong relative lift in sales and tickets versus the prior 90-day window, with moderate outperformance versus the same period last year. Average basket remains comparatively flat, indicating gains are primarily volume-driven. Impact: board reporting can distinguish traffic-led growth from ticket-size expansion when setting near-term actions.



Live Trend Analytics: Trailing 90-Day vs Same Period Last Year. Indexed + jittered curves preserve direction and comparison structure while obfuscating exact values.

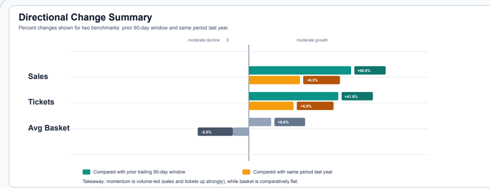


Fig 6. Directional change summary. vs prior 90D: Sales **+50.8%**, Tickets **+41.5%**, Avg Basket **+6.6%**. vs same period last year: Sales **+6.3%**, Tickets **+6.8%**, Avg Basket **-0.5%**.

Interpretation. This answers the analysis component of the research question: performance lift is mostly volume-led (sales and tickets), while average basket is comparatively stable. Operationally, this supports actions that focus on traffic and visit frequency rather than pricing changes.

Reproducibility & Ethics

No raw sensitive financial records on this poster. Secrets managed via environment variables and deployment secrets (not in git). Claims are bounded to validated data windows and documented coverage limits. Governance: `docs/legal/privacy-policy.md`, `docs/legal/eula.md`, `docs/legal/intuit-app-review-answers.md`.

References & Implementation

Architecture: ADR-0001 (workflow) · ADR-0002 (pipeline) · ADR-0004 (Lightspeed OAuth). KPI semantics: `docs/executive-analytics-grafana/kpi-dictionary.md`. Runbooks: `docs/runbooks/`. Dashboard: spokes-grafana.fly.dev. Figures generated: `python src/generate_figures.py`.